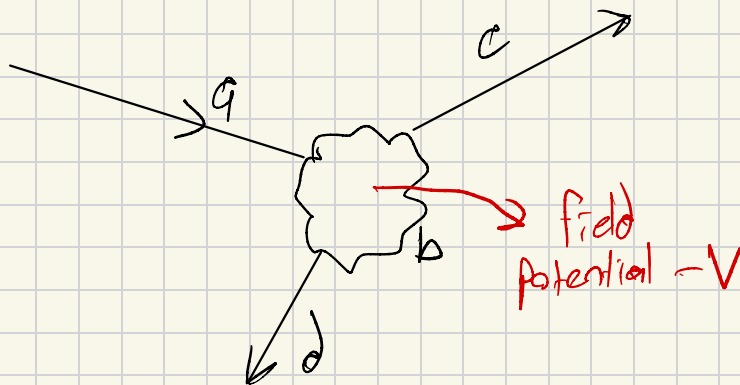




4) Interaction and fields

$$a + b \rightarrow c + d$$

4.1) Classical View



The potential V is a solution of the (static) Klein-Gordon eq.

$$\vec{\nabla}^2 V(\vec{R}) - \frac{1}{R_0^2} V(\vec{R}) = \delta(\vec{R})$$

R_0 : length of the interaction

Case -1: $R_0 = \infty$ (EM)

Solve $\vec{\nabla}^2 V(\vec{R}) = \delta(\vec{R})$ $V(\vec{R}) \propto \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$

Case -2: R_0 is finite (Strong interaction)

The potential is of the form $V(\vec{R}) \propto \frac{e^{-R/R_0}}{4\pi R}$

Yukawa potential (Nuclear Phys)

Heisenberg principle

$$\Delta E \Delta t \simeq \hbar$$

$$\Delta E = mc^2 \quad \Delta t \simeq \frac{\hbar}{mc^2}$$

$$R_0 \simeq c \Delta t = \frac{\hbar c}{mc^2} \sim \hbar c \simeq 200 \text{ MeV} \cdot \text{fm}$$

$$R_0 = 1 \text{ fm} \Rightarrow mc^2 \simeq 200 \text{ MeV}$$

1935
1947 discovery of π meson
 $m_\pi \simeq 140 \text{ MeV}$

The scattering amplitude of a particle in a potential $V(\vec{R})$ is given by the F.T. (Fourier transform) of the potential

$$M = g_a \int V(\vec{R}) e^{i\vec{q} \cdot \vec{R}} d\vec{R} \quad \text{with} \quad V(\vec{R}) = g_b \frac{e^{-MR}}{4\pi R} \quad R/R_0 \equiv MR$$

Where g_a is the coupling constant of particle a with $V(\vec{R})$

g_b is the constant related to particle b

$$\vec{q} = \vec{p}_a - \vec{p}_c \quad \text{transferred momentum}$$

the calculation gives
$$M = \frac{g_a g_b}{|\vec{q}|^2 + M^2}$$

This description works for low-energy interactions (Nuclear physics)

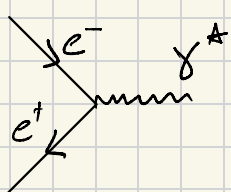
→ fails for higher energy processes:

- non-relativistic
- not Lorentz-invariant
- too simplistic view of the exchanged particle

4.2) Interaction by particle exchange

The correct framework to describe the interaction is QFT

- Preliminary remark



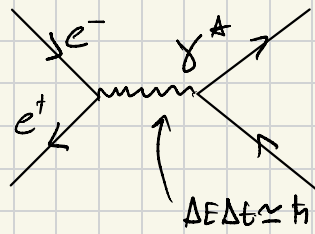
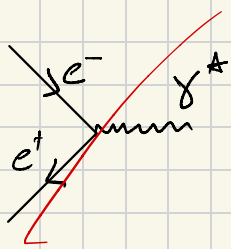
Energy conservation :

$$E_{e^+e^-} = E_+ + E_- = \sqrt{m_e^2 c^4 + p_+^2 c^2} + \sqrt{m_e^2 c^4 + p_-^2 c^2}$$

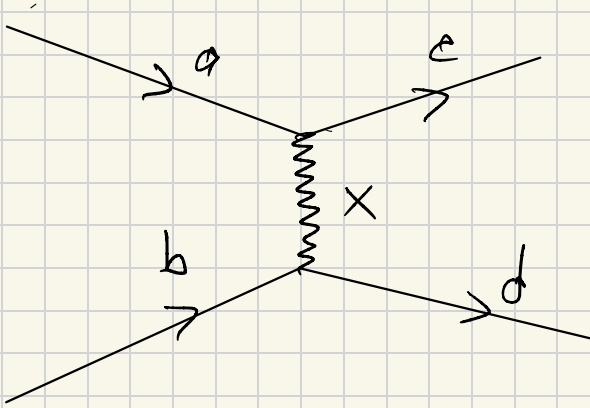
final state : photon $E_\gamma = p_\gamma c$, $\vec{p}_\gamma = \vec{p}_+ + \vec{p}_-$

We note $E_{e^+e^-} \neq E_\gamma$, The photon is virtual

$$E_{\gamma^*}^2 \simeq m_{\gamma^*}^2 c^4 + p_{\gamma^*}^2 c^2$$



o the scattering $a+b \rightarrow c+d$ can be described by a F.D.



X : Virtual particle that mediates the interaction.

Example: For a spin-0 particle X , it can be shown that the matrix element is:

$$M_{fi} = \frac{g_a g_b}{q^2 - m_x^2}$$

with $q = p_a - p_c = p_d - p_b$
4-momentum of the exchanged particle X

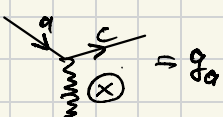
X is virtual hence $q^2 \neq m_x^2$

the particle is off-shell

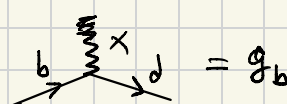


generally the matrix-element is composed of 3 parts

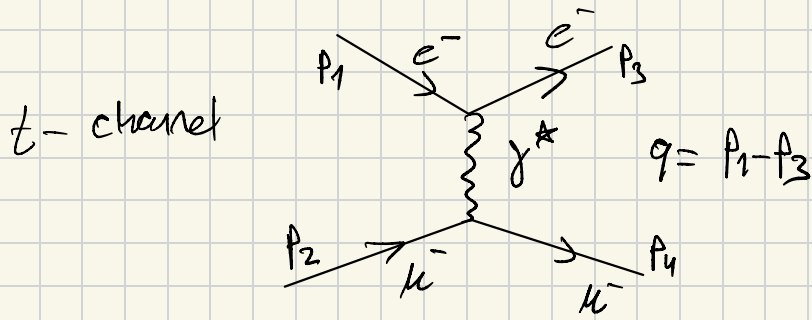
$$M_{fi} = \langle \psi_c | V_x | \psi_a \rangle \underbrace{\frac{1}{q^2 - m_x^2}}_{\text{Propagation of } X} \langle \psi_d | V_x | \psi_b \rangle$$



Propagation of X



4.3) Feynman rules for QED



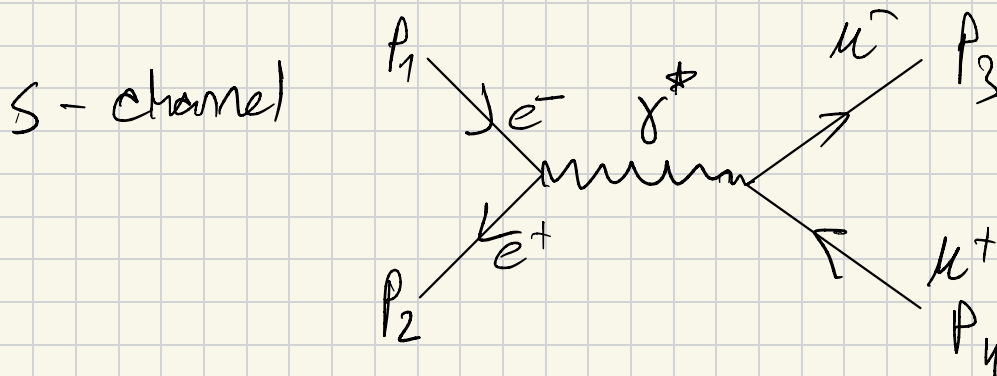
$$-iM = \left[\bar{u}(p_3) \cdot i e \overset{\text{charge}}{\gamma}^\mu u(p_1) \right] \left(\frac{-i g_{\mu\nu}}{q^2} \right) \left[\bar{u}(p_4) i e \gamma^\nu u(p_2) \right]$$

$$\bar{\psi} = \psi^\dagger \gamma^0 = (\psi^\dagger)^T \gamma^0$$

for t-channel

$$q^2 = E^2 - p^2 < 0$$

Space-like interaction



$$-iM = \left[\bar{v}(p_2) i e \gamma^\mu u(p_1) \right] \left(\frac{-i g_{\mu\nu}}{q^2} \right) \left[\bar{u}(p_3) i e \gamma^\nu v(p_4) \right]$$

s-channel $q^2 = E^2 - p^2 > 0$ time-like exchanged particle

$$\sigma \propto |M|^2$$

5) Interaction Strength

EM interaction

from atomic physics
 $\alpha \approx 1/137$

$M \propto e^2$ with introducing $\alpha = \frac{e^2}{4\pi\hbar c \epsilon_0}$ coupling constant

for a scattering $e^+e^- \rightarrow \mu^+\mu^-$ $\sigma \propto |M|^2 \propto \frac{e^4}{q^4} \approx \frac{\alpha^2}{q}$

Strong interaction

Similarities with QED because of its massless mediation $M_{\text{gluon}} = 0$

We introduce $\alpha_s = \frac{g_s^2}{4\pi}$ g_s color charge

We consider these 2 processes

$\overset{\text{SD}}{K^-} + \overset{\text{UUD}}{p} \rightarrow \overset{\text{Uds}}{\Sigma^0(1385)} \rightarrow \overset{\text{Uds}}{\Lambda} + \overset{\text{UUD}}{\pi^0}$

Strong interaction
 $\Gamma = 36 \text{ MeV}$
 $\tau = \frac{1}{\Gamma} \approx 10^{-23} \text{ s}$

Let's compare to a similar EM process

$\overset{\text{SD}}{K^-} + \overset{\text{UUD}}{p} \rightarrow \overset{\text{Uds}}{\Sigma^0(1192)} \rightarrow \overset{\text{Uds}}{\Lambda} + \gamma$ $\tau_{\text{EM}} \approx 10^{-19} \text{ s}$

$$\frac{\tau_{\text{EM}}}{\tau_s} = \frac{\Gamma_s}{\Gamma_{\text{EM}}} \approx \frac{\alpha_s^2}{\alpha^2}$$

$$\tau \propto \frac{1}{\Gamma} \Rightarrow \frac{\alpha_s}{\alpha} = \sqrt{\frac{\Gamma_{\text{EM}}}{\Gamma_s}} \approx 100$$

$$\Gamma \propto |M|^2 \approx \alpha^2 \Rightarrow \alpha_s = 100\alpha \approx 1$$

Weak interaction

- implies change of flavour and/or neutrino

$$\Sigma^- \rightarrow \Lambda + \pi^- \quad \tau \simeq 10^{-10} \text{ s} \quad \text{Weak}$$

$$\Sigma^0 \rightarrow \Lambda + \gamma \quad \tau \simeq 10^{-19} \text{ s} \quad \text{EM}$$

- Weak interaction implies massive boson propagation

$$\begin{array}{ccc} Z^0 & , & W^\pm \\ \parallel & & \parallel \\ 91 \text{ GeV} & & 80 \text{ GeV} \end{array}$$

Suppressed by $\frac{1}{M_W^2}$

Weak interaction is related to the Fermi const G_F

$$G_F = \frac{\sqrt{2}}{8} \cdot \frac{g_W^2}{\underbrace{M_W^2}} \simeq 1,16 \cdot 10^{-5} \text{ GeV}^{-2}$$

$$g_W : \text{weak charge} \quad \simeq 80 \text{ GeV} \Rightarrow g_W \simeq 0,65$$

$$\alpha_W = \frac{g_W^2}{4\pi} \quad \alpha_W \simeq \frac{1}{30}$$

$$\alpha \simeq \frac{1}{137}$$

$$\alpha_S \simeq 1$$

$$\alpha_W \simeq \frac{1}{30}$$